DE LA SALLE UNIVERSITY-DASMARIÑAS
COLLEGE OF ENGINEERING, ARCHITECTURE, AND TECHNOLOGY
ENGINEERING DEPARTMENT
COMPUTER ENGINEERING PROGRAM

SOFTWARE DESIGN

(T-CPET3228LA)

Enabling Assessment 2 – Log-in and Dashboard

Grade

Submitted By:
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Submitted To:

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Professor

ASSESSMENT TASK:

1. login and dashboard with database Version 2

2. Chapter 1 and II

A. Give all the screenshots of your dashboard and login

B. Explains each screenshot

Rubrics

Criteria	Functionality and Completeness
Dashboard	40%
Login	40
Document	20%
Total	100%

BarangayKonek: Design and Development of a Barangay-Centered Social Networking Platform for Community Engagement with E-Services

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CPE32

CHAPTER 1

INTRODUCTION

This chapter presents the background of the study and explains the problem that the research aims to address. It also includes the objectives of the study, the scope and limitations, the significance of the study, and the conceptual framework that guides the development of the proposed system.

1.1 Background of the Study

Technology-mediated communication has reshaped how individuals and communities interact, share information, and participate in governance processes. Online platforms and e-government services are widely used to support citizen engagement, improve transparency, and provide efficient access to public services (Bangero, 2025). In the Philippines, barangays the smallest political units of local governance serve as the frontlines of public service delivery and community interaction. However, many barangay operations remain heavily reliant on traditional or manual systems, which often hinder timely communication and active participation from residents (Alvero et al., 2025).

Several studies have highlighted the potential benefits of digital systems in improving barangay services. For example, digital information systems have been shown to streamline administrative functions such as resident profiling, record-keeping, and certificate issuance (Digital Barangay study, 2025). Digitizing governance processes also enhances efficiency, transparency, and service delivery, contributing to stronger local governance structures (Aliling et al., 2025).

Moreover, community participation in governance is significantly influenced by effective communication channels and information accessibility. A descriptive study on ICT use in a Philippine barangay found that residents often use gadgets and digital devices, yet are less updated on local events due to limited information flow (Alvero et al., 2025). Yet current digital government practices often rely on generic platforms such as social media which are not officially monitored or tailored to barangay needs leading to fragmented communication (Bangero, 2025).

A digital public square an interactive community space where residents can share ideas and engage in dialogue reinforces the notion that community engagement improves when communication is localized and centered on shared interests (Digital public square concept). By designing an integrated platform that allows not only access to e-services but also community interaction, barangay governance can transform from one-way announcements to participatory communication and collaboration.

BarangayKonek addresses these gaps by providing a barangay-centered social networking platform that integrates service access, resident interaction, and official communication in a single digital environment. This will enable residents to post updates, engage in discussions, and receive official announcements fostering stronger community engagement and collaboration.

1.2 Statement of the Problem

Absence of an official community interaction space. Residents currently depend on unofficial external social platforms that are not managed by barangay officials and lack contextual relevance to barangay affairs (Bangero, 2025).

Fragmented communication processes. Important announcements and information reach residents inconsistently due to lack of a centralized digital communication channel (Alvero et al., 2025).

Limited resident engagement. ICT use by residents does not fully translate to participation in barangay governance due to inadequate platforms catering to local dialogue and feedback (Alvero et al., 2025).

Disconnected e-services and social interaction. Existing systems focus on administrative transactions but do not facilitate interactive communication or community engagement

1.3 Objectives of the Study

1.3.1 General Objective

- To design and develop BarangayKonek, a barangay-centered social networking platform that enhances community engagement through interactive communication and integrated digital services.

1.3.2 Specific Objectives

1. To Develop a barangay web application that contains the following modules:
    a. Resident Management Module
    b. Document Request Module
    c. Payment Module
    d. Announcement Module
    e. Concern Reporting Module
    f. Community Feed Module
    g. User Management Module
    h. Activity Audit Log Module
    i. Reports Module
    j. Authentication and Security Module
2. To design the frontend and backend using the following languages and tools:
    a. HTML
    b. PHP
    c. MYSQL
    d. CSS
    e. Bootstrap
    f. XAMP
3. To implement system security through password hashing techniques.
4. To generate activity audit logs for monitoring and tracking user actions within the system.
5. To test and evaluate the system based on ISO software quality standards

1.4 Scope and Limitations

Scope:

1. Resident verification and profile management system.
2. Community feed that allows residents to post updates, comment, and interact with other users.
3. Official announcement board for barangay notices and advisories.
4. Document request module for services such as barangay clearance, certificate of residency, and other related documents.
5. Payment module for processing document request fees.
6. Concern reporting module where residents can submit complaints or community concerns.
7. Activity audit log module for monitoring user activities within the system.
8. Administrative dashboards for residents, barangay staff, and system administrators.
9. Reports module for generating records and system activity summaries. Authentication and security module using password hashing techniques

Limitations:

1. The implementation of the system is limited to one barangay for development and testing purposes.
2. The system is developed as a web-based application, and mobile application development is not included in the scope of this study.
3. Private messaging between residents is not included and the system focuses primarily on community interaction through the community feed.
4. The payment module supports basic transaction recording and does not integrate with external online payment gateways.
5. System evaluation will be conducted based on ISO software quality standards, focusing on usability, reliability, and functionality.

1.5 Significance of the Study

This study on the development of Barangay-Centered Social Networking Platform for Community Engagement is expected to be significant and beneficial to the following:

Residents. Provides an officially moderated online space to interact, access services, and participate in local discussions.

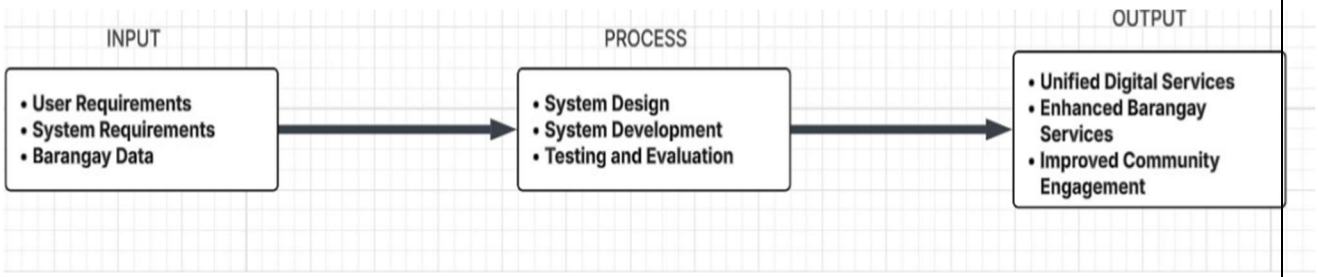
Barangay Officials. Improves communication efficiency, transparency, and responsiveness to community needs.

Local Government Units. Offers a replicable model for digital governance that strengthens citizen engagement.

Researchers and Developers. Contributes to the body of knowledge on integrating social networking with e-government service

1.6 Conceptual Framework

The conceptual framework of this study is based on the idea that community engagement increases when residents are provided with accessible communication channels, reliable information sources, and integrated digital services. BarangayKonek is designed to combine these three elements into one unified platform



1.7 Definition of Terms

Community Engagement — Active participation of residents in communication and governance

processes facilitated by digital platforms.

E-Governance — Use of ICTs to improve government operations and public services.

Resident Profile — Verified account associated with a barangay household.

Social Networking Platform — A digital medium that enables user interaction, content sharing, and community building**.**

Chapter 2

This chapter discusses the related literature and studies that support the development of the proposed system. It presents information about digital governance, community engagement through digital platforms, and digital systems used in local government. The chapter also identifies the research gap that the study aims to address.

2.1 Digital Governance and E-Government Systems

Digital governance refers to the use of information and communication technologies to support government operations and improve service delivery to citizens. According to the United Nations (2022), digital government platforms enhance transparency, efficiency, and accessibility of public services. These systems allow citizens to access information and services through online portals, reducing the need for physical visits to government offices.

Similarly, the World Bank (2022) states that digital transformation in the public sector improves administrative efficiency and strengthens the relationship between governments and citizens. Through digital platforms, governments can streamline services such as documentation, information dissemination, and citizen feedback.

Studies conducted in the Philippines also show that the adoption of digital systems in local government units improves administrative processes and communication with residents. Santos and Cruz (2023) found that ICT-based platforms allow local government offices to manage resident records more efficiently and provide faster service delivery.

However, many digital government systems are primarily designed for administrative functions and often lack features that encourage active community participation.

2.2 Community Engagement through Digital Platforms

Community engagement refers to the involvement of residents in discussions, activities, and decision-making processes that affect their local community. Digital platforms have become important tools for encouraging participation by allowing citizens to share information and communicate with others.

Research shows that online platforms can strengthen civic participation by providing accessible communication channels for citizens. Kumar and Nanda (2024) explain that digital platforms allow citizens to interact with government institutions and other community members, enabling them to share opinions, report concerns, and participate in discussions about public issues.

Similarly, Garcia and Ramirez (2024) state that digital community platforms encourage collaboration among residents and improve communication between local governments and citizens. These platforms provide spaces where residents can exchange ideas and stay informed about community developments.

Despite these benefits, many communities rely on general social media platforms for local communication. These platforms are not specifically designed for governance and often lack identity verification, official moderation, and integration with government services.

2.3 Social Networking Platforms for Local Communities

Social networking platforms enable users to create profiles, share information, and interact with other members of a digital community. These platforms have become widely used tools for communication and information exchange.

According to OECD (2023), digital communication platforms facilitate collaboration and engagement by enabling citizens to interact with each other and with government institutions. By allowing users to share posts, comments, and feedback, social networking platforms support the exchange of ideas and information within communities.

Localized social networking systems can also strengthen community relationships by providing platforms where residents discuss issues affecting their locality. Through these platforms, residents can interact with one another, share updates, and participate in community activities.

However, traditional social media platforms may not be suitable for local governance because they lack formal structures for managing community information and government services.

2.4 Digital Systems for Barangay Administration

Several studies have explored the use of digital systems to improve barangay management and service delivery. These systems are designed to streamline administrative processes and improve communication between barangay officials and residents.

For example, Alvero et al. (2025) found that ICT-based barangay management systems improve record organization and service processing. These systems enable barangay offices to maintain resident databases, generate reports, and manage document requests efficiently.

Similarly, Aliling et al. (2025) reported that digital governance initiatives at the community level enhance transparency and improve communication between residents and barangay officials. By digitizing administrative processes, barangays can provide faster services and maintain better records.

Although these systems improve administrative efficiency, many of them do not include interactive communication features that allow residents to participate actively in community discussions.

References

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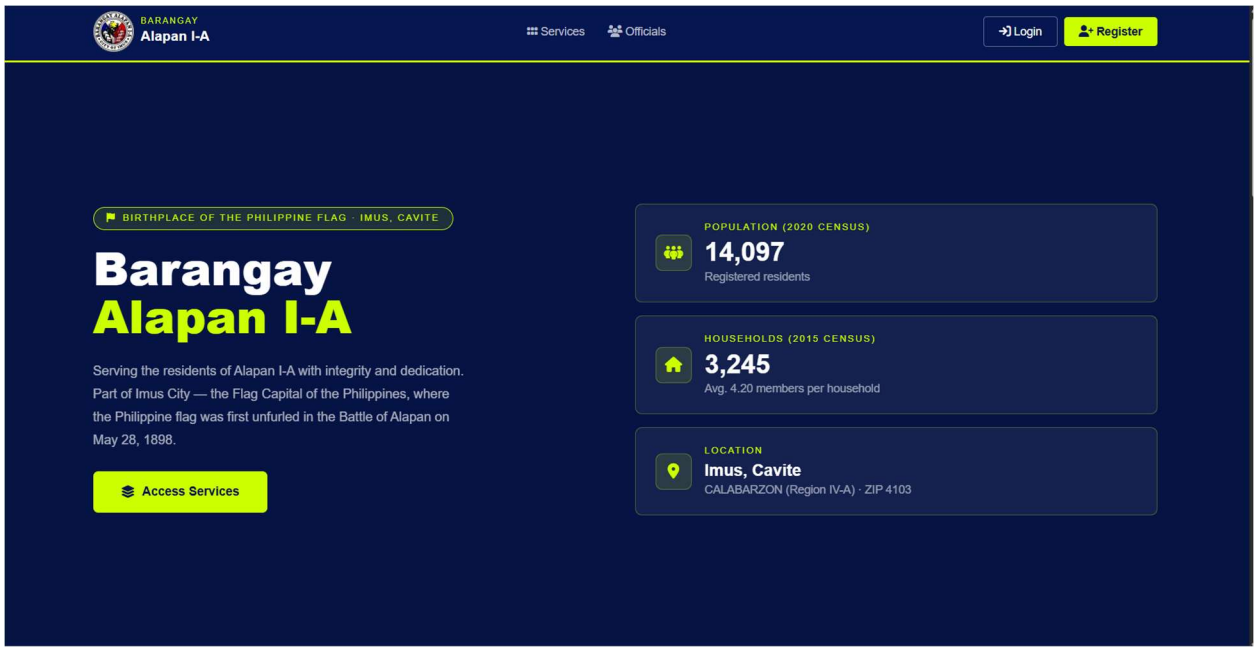
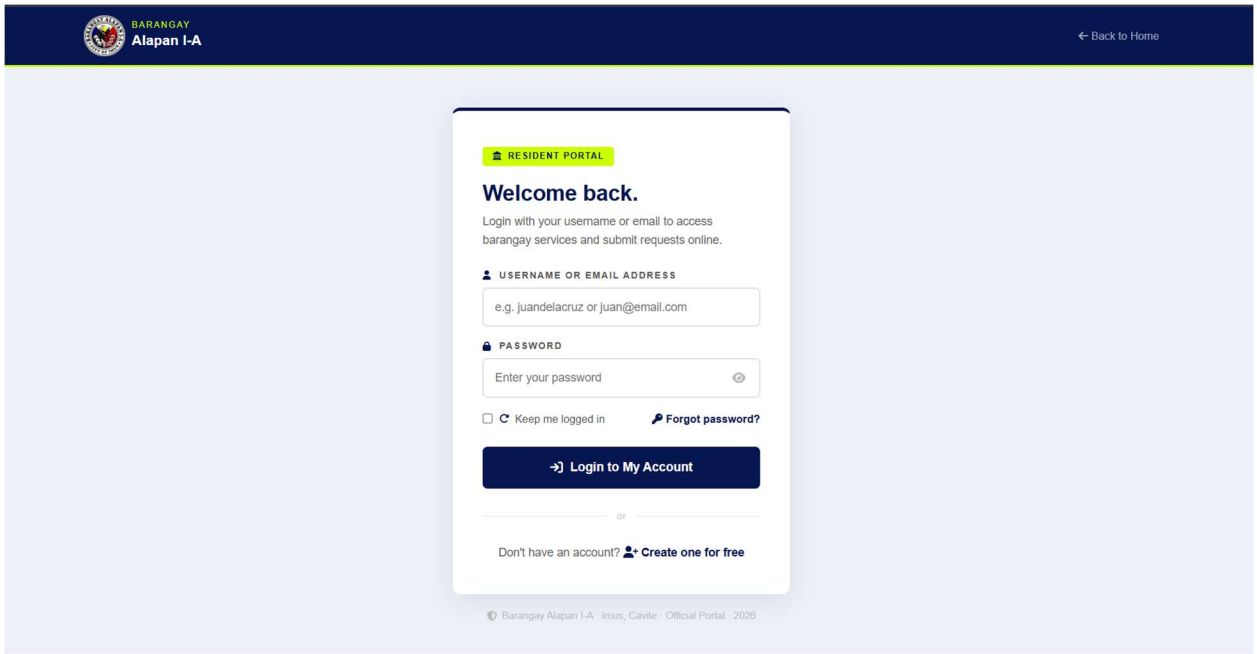
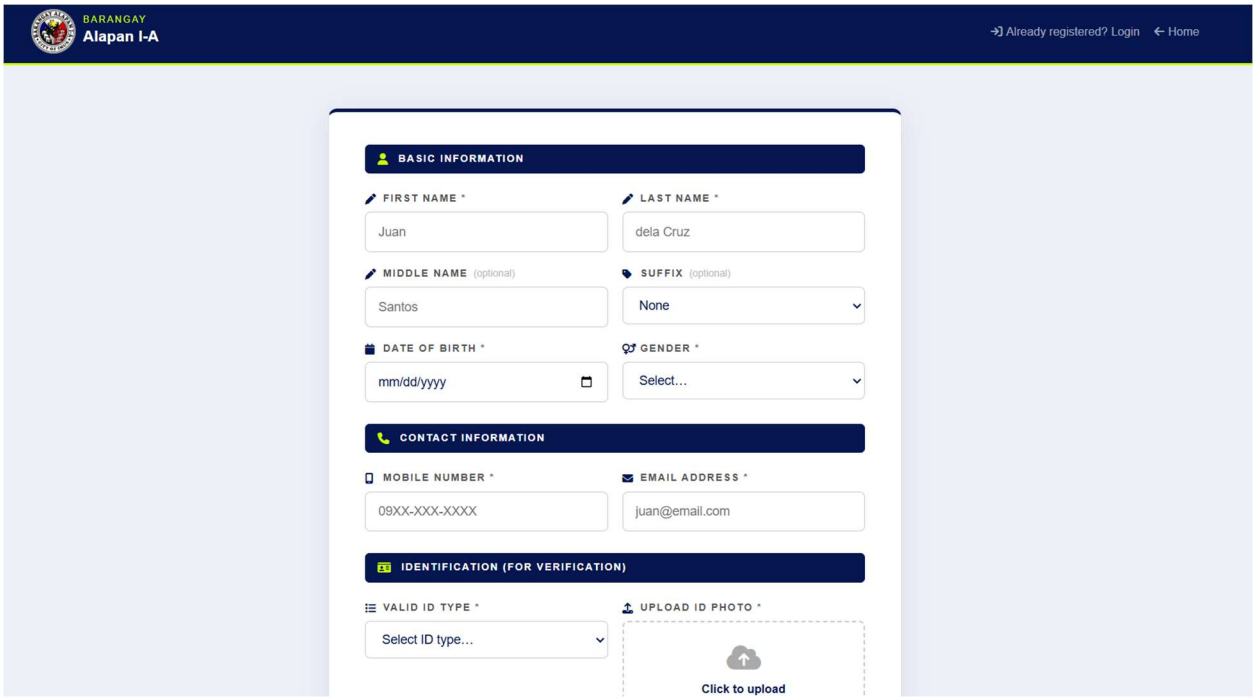
BarangayKonek: Screenshots



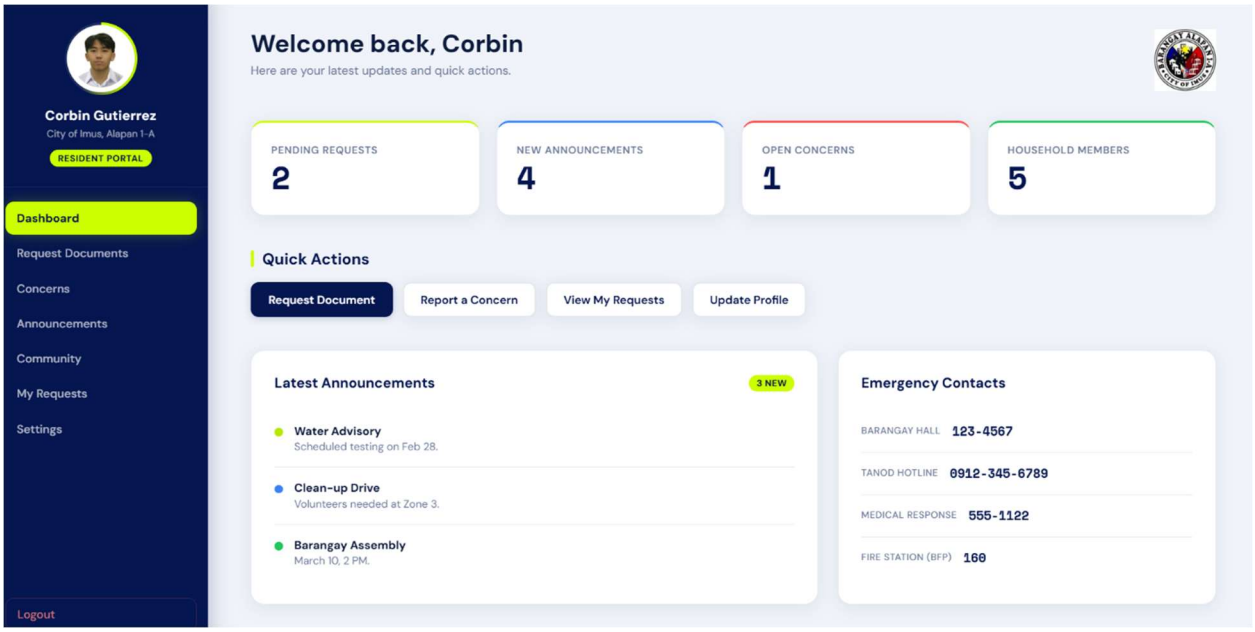
Home Page: This is where the overview of the BarangayKonek Website



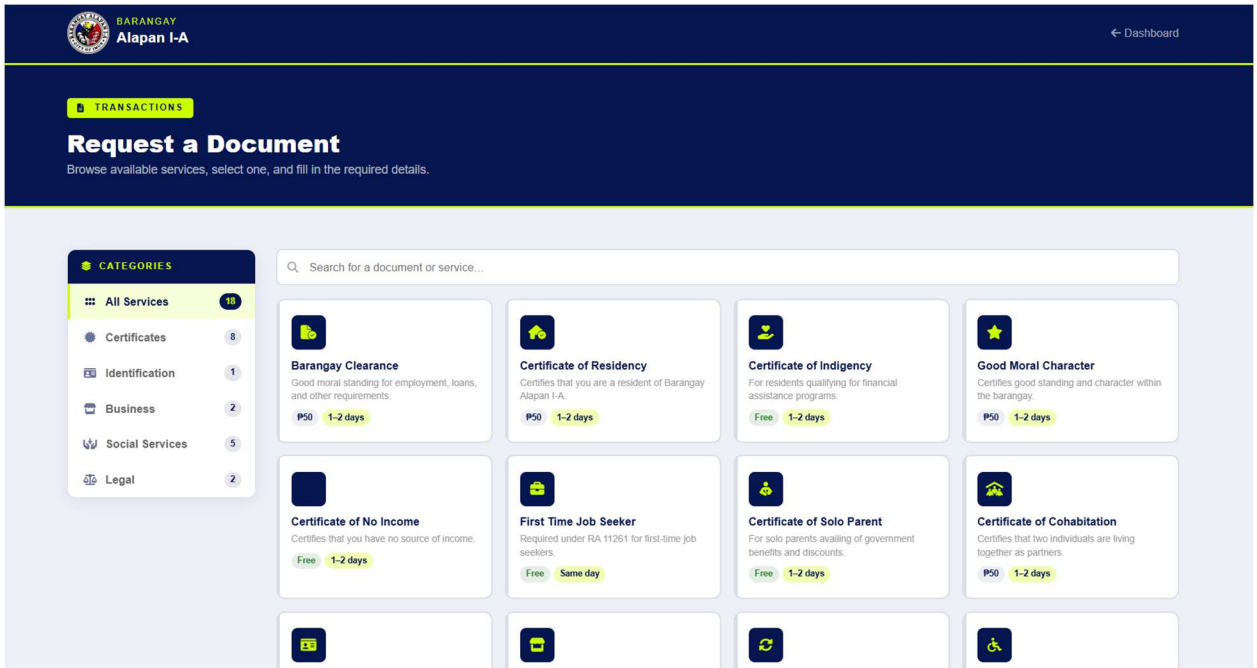
Login Page: This is where the users will log their accounts. It have Forgot Password and Create Password



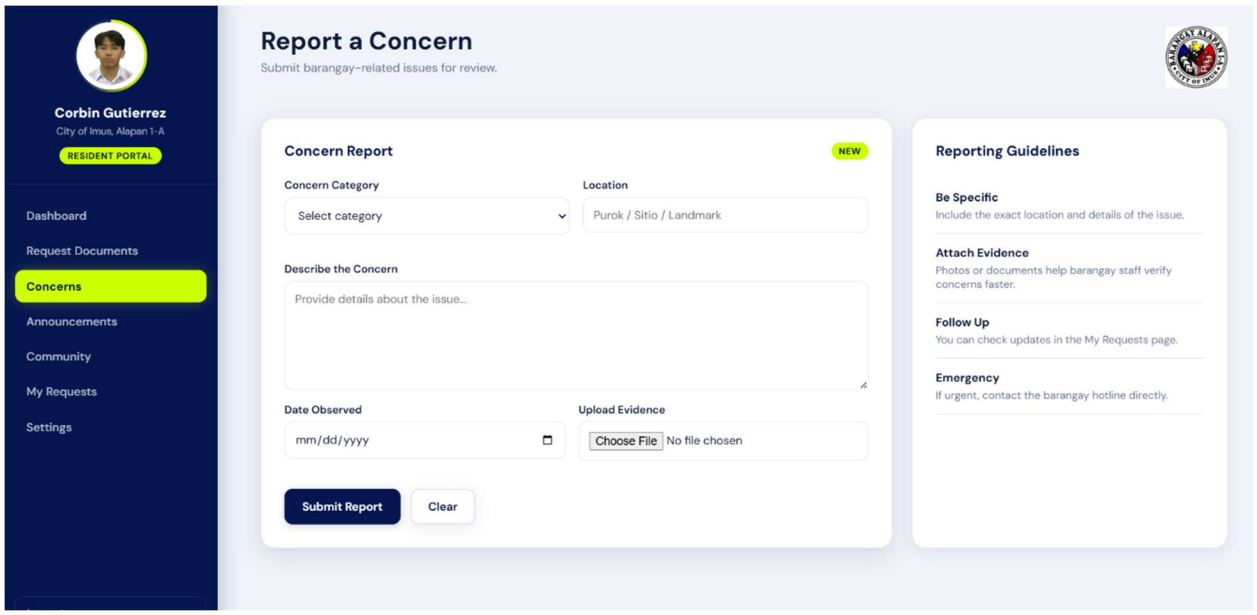
Registration Page: This is where the user will log their credentials with File Upload feature



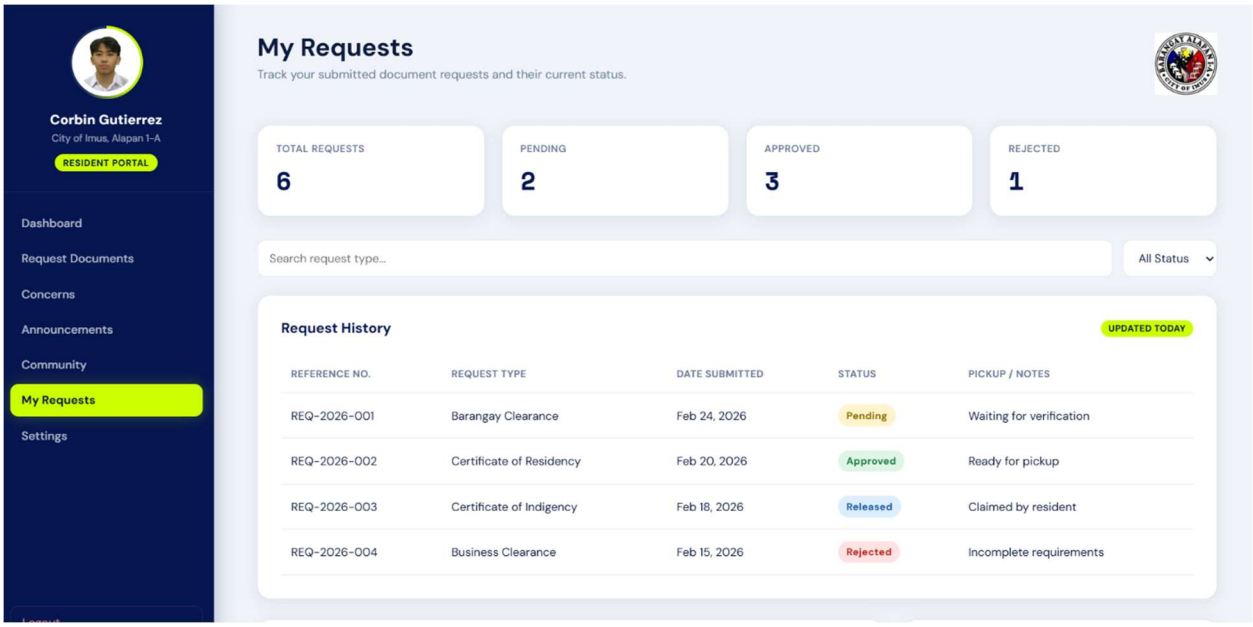
Resident Dashboard: This is the Resident Dashboard. There are multiple features in the dashboard and the functions are included in the side panel with cards for quick overview.



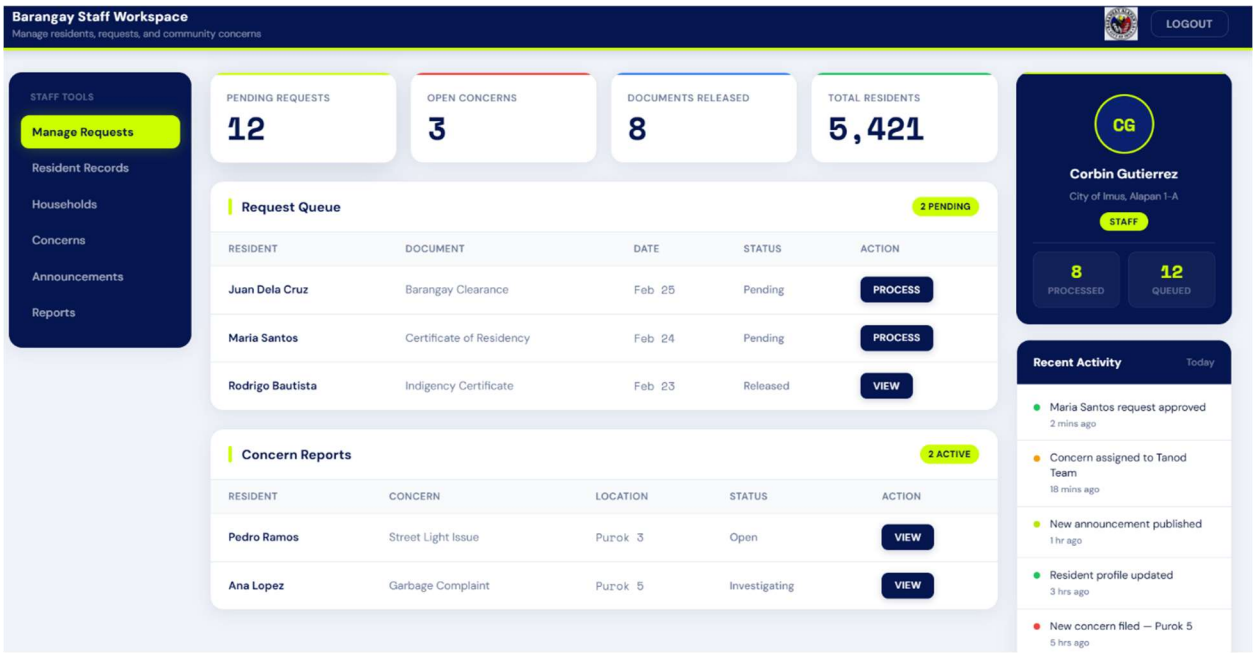
Request Page: This is where the user will request specific documents to the staffs.



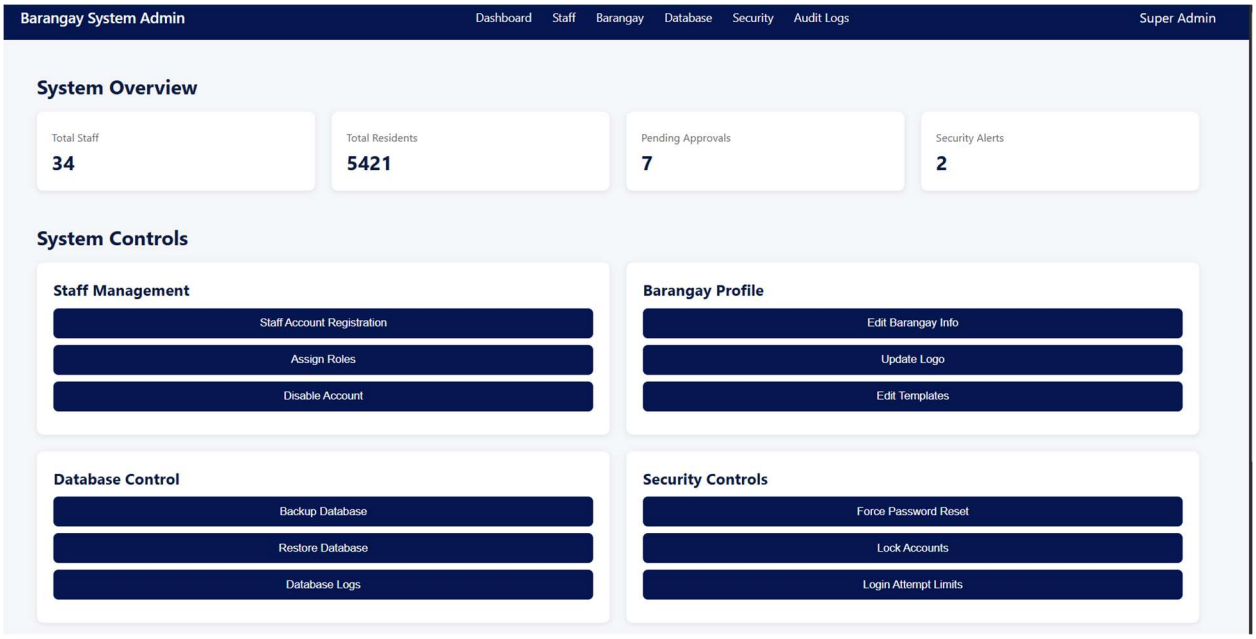
Concern Page: This is a form page where the residents will file their concerns.



My Requests Page: This is where the resident user will see their number of request, pending request, approved request, and etc.



Staff Dashboard Page: This is the dashboard for staffs. This is the quick overview of their functions, and they can see their requests queue and concern reports



Super Admin Dashboard: This is the dashboard for super admin. This is the quick overview of their functions, and it has a simplified design, they have control over database, staff, security and barangay profiles.